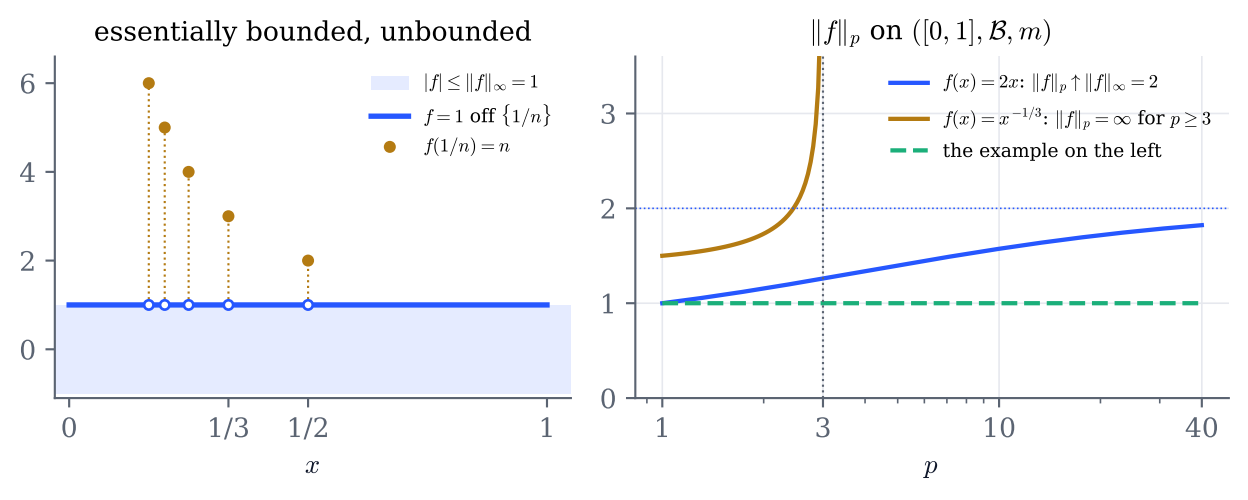
essentially bounded, unbounded
$|f| \le \|f\|_\infty = 1$
$f = 1$ off $\{1/n\}$
$f(1/n) = n$
6
4
2
0
0
1/3
1/2
1
x
$\|f\|_p$ on $([0,1], \mathcal{B}, m)$
$f(x) = 2x$: $\|f\|_p \uparrow \|f\|_\infty = 2$
$f(x) = x^{-1/3}$: $\|f\|_p = \infty$ for $p \ge 3$
the example on the left
3
2
1
0
1
3
10
40
p